

New Community Garden Site for Orange County, Florida Community

Taylor Beebe[1] and Michael Gilbrook[2]
Department of Interdisciplinary Studies[1], University of Central Florida



Introduction

As defined by the United States Department of Agriculture (USDA), community gardens "are collaborative projects on shared open spaces where participants share in the maintenance and products of the garden, including healthful and affordable fresh fruits and vegetables" (2006). These collaborative community spaces are especially integral to communities impacted by crisis or disaster, poverty, and low-resource access. Community gardens can serve as "sanctuary and place for social exchange… [,] a source of food…[, and] a post-disaster learning space" (Brantz & Sharma, 2020). Orange County, Florida is home to various communities that could benefit from a community-led garden, as seen through various recent examples such as within the Parramore District. As of 2023, various Parramore community members and officials came together to establish a community garden within one Parramore District neighborhood; "This farm aims to provide easier access and more affordable fresh products to the Parramore community…" (Osorio, 2023). Orange County boundary and census data, provided by the Florida Geographic Data Library, and food access data, provided by the USDA Food Access Research Atlas, were utilized to designate a similar Orange County community that could benefit from a garden. After an area-of-interest (AOI) was designated, various factors were applied to the AOI to determine the best physical site for a garden within the community boundary. Garden-specific factors, such as light availability, water feature and roadway access, were determined from the Orange County Community Gardeners Guide (OCCGG). Various community gardens, such as the Parramore District's, and online guides were referenced for specific metrics (Berle & Westerfield, 2013; Cullheim, 2020). Other factors were determined to limit the AOI to areas with a functionally adjustable land use and align with future zoning designations. (Table 1.)

Factors	Low Suitability (1)	Moderate Suitability (2)	High Suitability (3)
Water Access	Greater than a 10-minute walk.	Between 5-10-minute walk.	Within a 5-minute walk.
Road Access	Greater than 200 ft. away.	Between 100-200 ft. away.	Less than 100 ft. away.
Hours of Light (GHI)	Within the lower 33% of overall average.	Between the upper and lower averages.	Within the upper 33% of overall average.
Land Cover	All other designations.	NA.	Disturbed or barren land.
Zoning	All other designations.	NA.	Agricultural designation.
Area Size	NA.	NA.	Greater than or equal to 9,900 sq. ft.
"Sacred Cows" (0 or 5)	Road, water, wetland, and utility areas.	N/A.	N/A.

Table 1. Suitability scale discerned from garden site factors.

Methods

Locating an AOI: An attribute query was utilized to designate the Orange County (OC) boundary. This boundary acted as a "cookie cutter," or clip, reducing census data to OC. OC census data received an initial attribute query, which located communities that make less than Florida's median household income ($73,311, within 2023), more than 50% of the population lives below the poverty line, and 33% of the population lives 1 mile (urban) and 20 miles (rural) from the nearest supermarket (U.S. Census Bureau, 2023; U.S. Department of Agriculture, 2019). Of ten parcels located, one densely populated Apopka, Florida parcel ($43,542 MEDHHINC, 50.55% PCT_POV) did not currently have a community garden, leading to its selection as the AOI (Figure 2). **Locating a Suitable Garden Site:** All layers, excluding water features, were clipped by the AOI. (Many water features existed near, but not within the AOI, which could contribute as a water source.) Various features generated a (A) suitability layer to be averaged, and a (B) 'sacred cow' layer that excluded features below unsuitable. (A, water) Water features received a 5-minute (1320 ft.) and 10-minute (2640 ft.) buffer, which were converted to a raster format, utilizing the distance field, and reclassified to the 1-3 suitability scale. (B, water) Water features were converted to a raster format, and reclassified as either 0 (water features) or 5 (all other areas). This method was utilized to exclude any features deemed below unsuitable, or 0, values from final computations. (A, roads) Road centerlines received a 25 ft. buffer to account for actual full road width. Full roads then received a 100 ft. and 200 ft. buffer, were converted to a raster format, utilizing the distance field, and reclassified to the 1-3 suitability scale. (B, roads) Full roads were converted to a raster format, reclassified as either 0 (roads) or 5 (all other areas) to exclude roads from final computations. Land cover and zoning features were both utilized to account for physical ground cover and above ground building designations. (A, land cover) Land cover data was converted to a raster format, utilizing the land cover description field, and reclassified to the 1-3 suitability scale. (B, land cover) Land cover raster data was reclassified as either 0 (wetlands and utilities) or 5 (all other areas). (A, zoning) Zoning features were converted to a raster format, utilizing the zoning field, and reclassified to the 1-3 suitability scale. Areas of agricultural use were highlighted. (A, light) The global horizontal irradiation (GHI) raster was reclassified to the 1-3 suitability scale, prioritizing areas of high light. All raster data following the (A) 1-3 suitability scale were averaged using the 'mean' cell statistic…

Results

Two parcels were selected among the four returned from the model (Figure 1). These two parcels were regularly shaped, allowing for ease of planning and future construction within located areas. **Location A** is located within the Northeast section of the AOI. This location has an area of ~36,939 sq. ft.; is located within 5 min. of water features and 100 ft. of roadways; receives moderate light; has a land cover (barren) and zoning (low density) designation that allows for the construction of a community garden. **Location B** is located within the Southwest section of the AOI. This location has an area of ~21,547 sq. ft.; is located within 5 min. of water features and 100 ft. of roadways; receives high light; has a land cover (low density) and zoning (agricultural) designation that allows for the construction of a community garden. All metrics are locations are outlined within Figure 2.

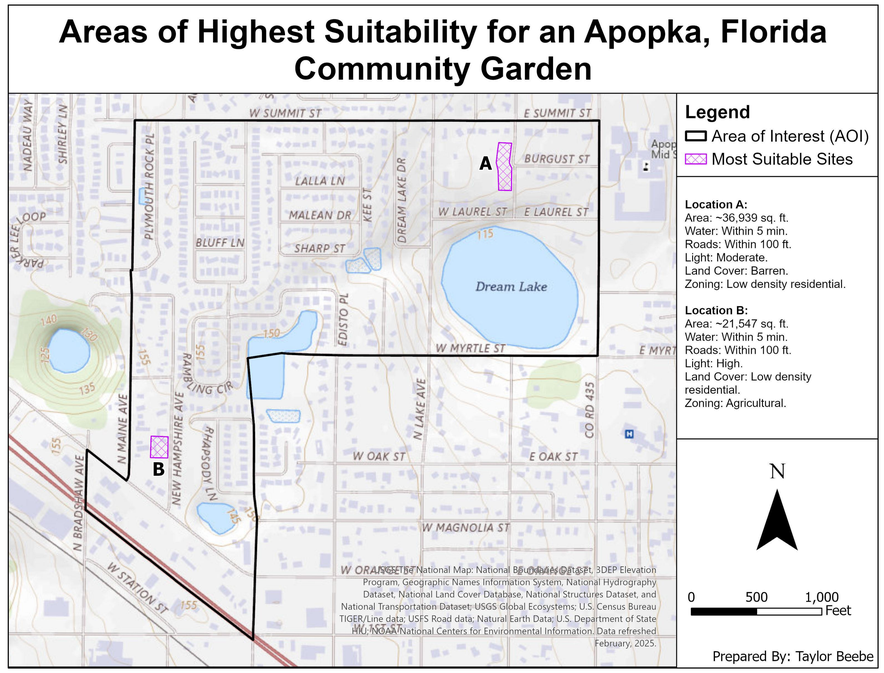


Figure 2. This image depicts two suitable locations for a garden within the Apopka, Florida AOI.

All (B) 'sacred cow,' or 0 or 5, raster data were combined using the 'minimum' cell statistic; areas below unsuitable, or 0, were highlighted in this layer (water, roads, wetlands, and utilities). Following this, the average suitability and 'sacred cow' layers were combined using the 'minimum' cell statistic, highlighting areas below unsuitable within the averaged data layer. This 'suitable sites' raster was then reclassified to limit data to sites of greater than or equal to ~2.3 suitability, converted to a polygon, and clipped by the AOI. Areas greater than or equal to 9,900 sq. ft. were selected via attribute query. All steps are shown in Figure 1. Two of four polygons were selected based on shape regularity, or squareness (Figure 2).

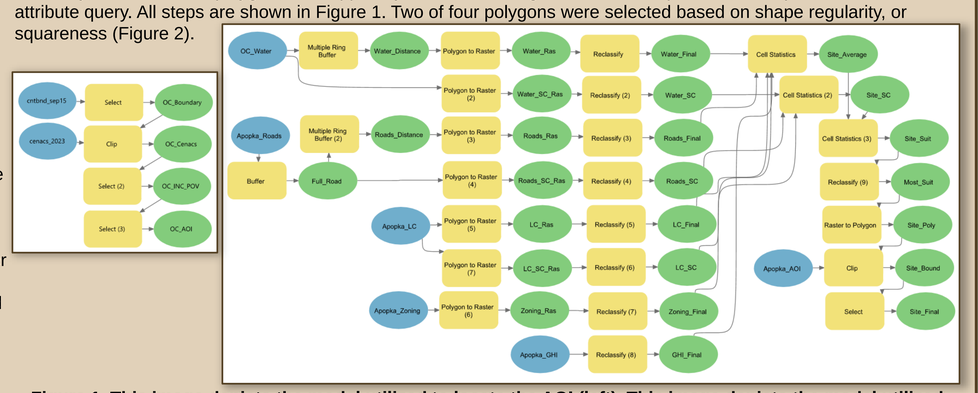


Figure 1. This image depicts the model utilized to locate the AOI (left). This image depicts the model utilized to generate final community garden locations within the AOI (right).

Discussion

Both sites identified provide plausible locations for a community-led garden, which could directly benefit local neighbors within the Apopka, Florida boundary. Both sites meet most of the suitability factors provided; however, certain criteria limit each parcel's plausibility for implementation. For example, although location A meets most of the suitability criteria (Table 1), average light received by the area is moderate. Furthermore, although location B meets most of the suitability criteria (Table 1), preliminary restoration of the area would need to occur, due to unkempt agricultural conditions of the current parcel. However, it is essential to consider the adaptability of community gardens. Community gardens can take a variety of forms and are not limited by the specific criteria determined within this study. For example, water can be collected through a variety of methods (rain barrels, nearby municipal utilities, etc.) and is not limited to nearby water features. After a location site is chosen, precise estimates would need to occur on site to deduce how gardens would appear and function for unique communities, such as within Apopka.



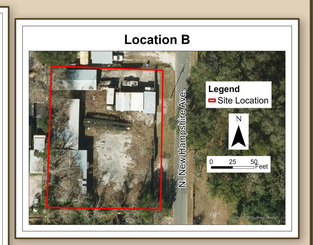


Figure 3. Aerial imagery of location A (left) and location B (right).

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Data Sources

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